



Gosford High School

Year 12

2012

Higher School Certificate

Mathematics

Assessment Task 3

Time Allowed – 60 minutes

(reading time 5 minutes)

Remember to use the **provided multiple choice answer sheet** to answer **Question 1**

Remember to start each new question on a **new page** for **Questions 2 to 5**

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

QUESTION 1 (please use the provided multiple choice answer sheet)**5 marks**

i) $e^{2\ln x} = ?$

A) $2 \ln x$

B) $2x$

C) 2^x

D) x^2

ii) The $\int x e^{x^2} dx$ is equal to

A) $\frac{1}{2} e^{x^2} + c$

B) $2e^{x^2} + c$

C) $\frac{e^{x^3}}{3} + c$

D) None of these

iii) The graph of $y = \cos\left(x + \frac{\pi}{2}\right)$ for all values of x is the same as the graph of :-

A) $y = \sin x$

B) $y = \cos x$

C) $y = -\sin x$

D) $y = -\cos x$

iv) If x is an extremely small positive value, which one of the following is NOT true

A) $\cos x \doteq 1$

B) $x \doteq \sin x \doteq \tan x$

C) $\sin x < x < \tan x$

D) $\tan x < x < \sin x$

v) If $A = B + Ce^{-kt}$ then

A) $t = \frac{1}{k} \log_e \left[\frac{A-B}{C} \right]$

B) $t = \frac{1}{k} \log_e \left[\frac{B-A}{C} \right]$

C) $t = \frac{1}{k} \log_e \left[\frac{C}{A-B} \right]$

D) $t = -\log_e \left[\frac{A-B}{kC} \right]$

QUESTION 2*start a new page***(11 marks)**

a) Find $\frac{d}{dx} [e^{x^2-2x+7}]$ (1)

b) Find primitives of (i) $\frac{1}{7-2x}$ (1)

(ii) e^{4-3x} (1)

c) Evaluate $\int_{-1}^1 \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$ (2)

d) If $y = \log_e \left(\frac{1+x}{1-x} \right)$ show that $\frac{dy}{dx} = \frac{2}{1-x^2}$ (3)

e) The portion of the curve $y = e^x + 2$ from $x = 0$ to $x = 1$ is rotated about the x axis.

Find the volume of the solid generated. (3)

QUESTION 3*start a new page***(10 marks)**

- a) Find $\int \sin(2x) dx$ (1)
- b) State the amplitude and period of the curve $y = -2 \cos\left(\frac{\pi x}{2}\right)$ (2)
- c) Find $\frac{d}{dx} [\cos(3x)]$ (1)
- d) If $y = \sin^2\left(\frac{x}{2}\right)$, find $\frac{dy}{dx}$ (1)
- e) Find the equation of the tangent at $x = \frac{\pi}{6}$ on the curve $y = \tan(2x)$, giving your answer in simplest exact form. (3)
- f) Consider the function $f(x) = 1 - 2 \sin\left(x + \frac{\pi}{4}\right)$ for $0 \leq x \leq 2\pi$
- (i) Without using calculus, state the maximum value of $f(x)$. (1)
- (ii) State the value of x , in the given domain, for which this maximum occurs. (1)

QUESTION 4*start a new page***(11 marks)**

- a) Find the value of $\log_7 43700$ correct to 3 significant figures. (1)
- b) Sketch the curve $y = \log_e(x + 2)$, carefully labelling the intercepts on the coordinate axes and any asymptotes. (2)
- c) Consider $f(x) = x - 3\ln x$
- Find the coordinates of the stationary point on the curve $y = f(x)$ and determine its nature. (4)
- d) Show that the curve $y = xe^{-x}$ has a point of inflexion at $x = 2$ (4)

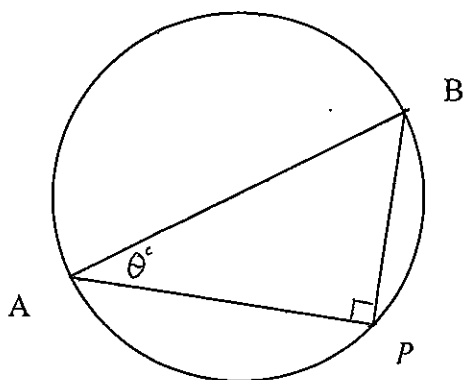
QUESTION 5*start a new page***(13 marks)**

- a) The gradient of the tangent at a point (x, y) on the curve $y = f(x)$ is given by

$\cos(2x) - \sec^2 x$. If the curve passes through the point $\left(\frac{\pi}{4}, \frac{1}{2}\right)$, find $f(x)$. (2)

- b) A point P moves along the circumference of a circle with radius R and diameter AB.

If $\angle APB = \frac{\pi^c}{2}$ and $\angle PAB = \theta^c$.



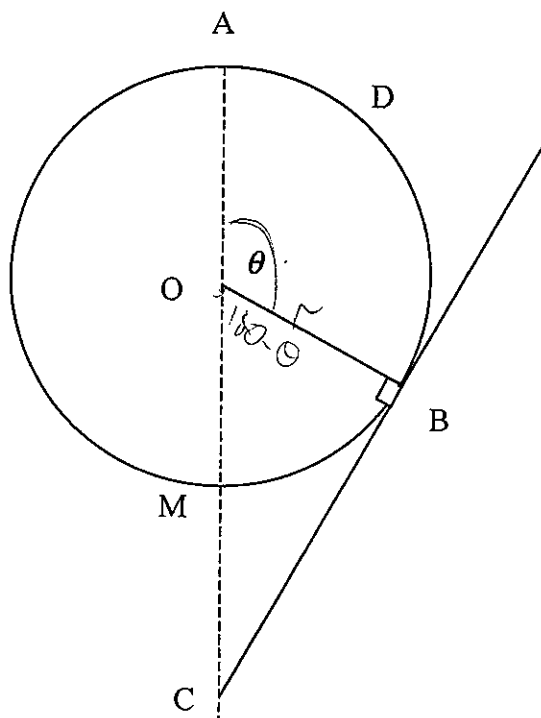
(i) Show that $PA = 2R \cos \theta$. (1)

(ii) Hence show that the area of $\triangle PAB = 2R^2 \sin \theta \cos \theta$ (1)

(iii) Given the identity that $2 \sin \theta \cos \theta = \sin 2\theta$, the above area (A) of $\triangle PAB$ can be written as $A = R^2 \sin 2\theta$.

Show that the area of $\triangle PAB$ is a maximum when $\theta = \frac{\pi^c}{4}$ (3)

c)



In the diagram, O is the centre of the circle with radius 'r'.

A, O, M and C are collinear. Given Arc ADB + interval BC = l

(i) Show that $\theta - \tan \theta = \frac{l}{r}$. (3)

(ii) If the area of the minor sector AOB is $\frac{3}{8}$ of the area of the circle,
find θ and ratio $\frac{l}{r}$. (3)

END OF EXAMINATION

MULTIPLE CHOICE ANSWER SHEET

YEAR 12 MATHEMATICS ASSESSMENT TASK 3

QUESTION 1

(i)	A	B	C	D
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(ii)	A	B	C	D
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(iii)	A	B	C	D
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(iv)	A	B	C	D
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(v)	A	B	C	D
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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

SOLUTIONSQuestion 1 (i) D (ii) A

(iii) C (iv) D

(v) C

Question 2 a) $(2x-2)e^{x^2-2x+7}$ (1)b) (i) $-\frac{1}{2} \log_e (7-2x) + C$ (1)(ii) $-\frac{1}{3} e^{4-3x} + C$ (1)

$$c) \int_{-1}^1 \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \left[\log_e (e^x + e^{-x}) \right]_{-1}^1 \quad (1)$$

$$= \log_e (e + e^{-1}) - \log_e (e^{-1} + e)$$

$$= 0 \quad (1)$$

$$d) y = \log_e (1+x) - \log_e (1-x)$$

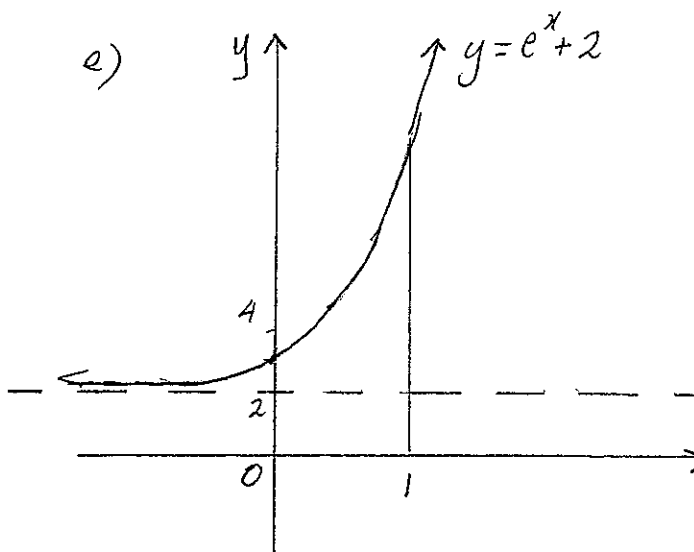
$$= \frac{1}{1+x} - \frac{-1}{1-x} \quad (2)$$

$$= \frac{1}{1+x} + \frac{1}{1-x}$$

$$= \frac{1(1-x) + 1(1+x)}{(1+x)(1-x)}$$

$$= \frac{2}{1-x^2} \quad (1)$$

e)



$$V = \pi \int_0^1 (e^x + 2)^2 dx \quad (1)$$

$$= \pi \int_0^1 (e^{2x} + 4e^x + 4) dx$$

$$= \pi \left[\frac{1}{2} e^{2x} + 4e^x + 4x \right]_0^1 \quad (1)$$

$$\therefore V = \pi \left[\frac{1}{2} e^2 + 4e + 4 - \left(\frac{1}{2} + 4 + 0 \right) \right]$$

$$= \pi \left[\frac{1}{2} e^2 + 4e - \frac{1}{2} \right] \text{ cubic units} \quad (1)$$

Question 3

a) $\int \sin(2x) dx = -\frac{1}{2} \cos(2x) + c \quad (1)$

b) Amplitude = 2, Period = $\frac{2\pi}{\pi/2}$

$$= 4 \quad (2)$$

c) $\frac{d}{dx} [\cos(3x)] = -3 \sin(3x)$

d) $y = \left[\sin\left(\frac{x}{2}\right) \right]^2$

$$\frac{dy}{dx} = 2 \left[\sin\left(\frac{x}{2}\right) \right]' \cdot \frac{1}{2} \cos\left(\frac{x}{2}\right) \quad (1)$$

$$= \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$$

$$e) \quad y = \tan(2x) \quad \text{when } x = \frac{\pi}{6}, \quad y = \tan\left(\frac{\pi}{3}\right) = \sqrt{3}$$

$$\frac{dy}{dx} = 2\sec^2(2x) \quad \text{when } x = \frac{\pi}{6}, \quad \frac{dy}{dx} = 2\sec^2\left(\frac{\pi}{3}\right) = 8 \quad (2)$$

$$\text{Equation of Tangent } y - \sqrt{3} = 8\left(x - \frac{\pi}{6}\right) \quad (1)$$

$$y = 8x - \frac{4\pi}{3} + \sqrt{3}$$

$$f) \quad (i) \quad -1 \leq \sin\left(x + \frac{\pi}{4}\right) \leq 1.$$

$$\therefore \text{Max. Value of } f(x) \text{ is } 3 \quad (1)$$

$$(ii) \quad \text{Solving } \sin\left(x + \frac{\pi}{4}\right) = -1$$

$$x + \frac{\pi}{4} = \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

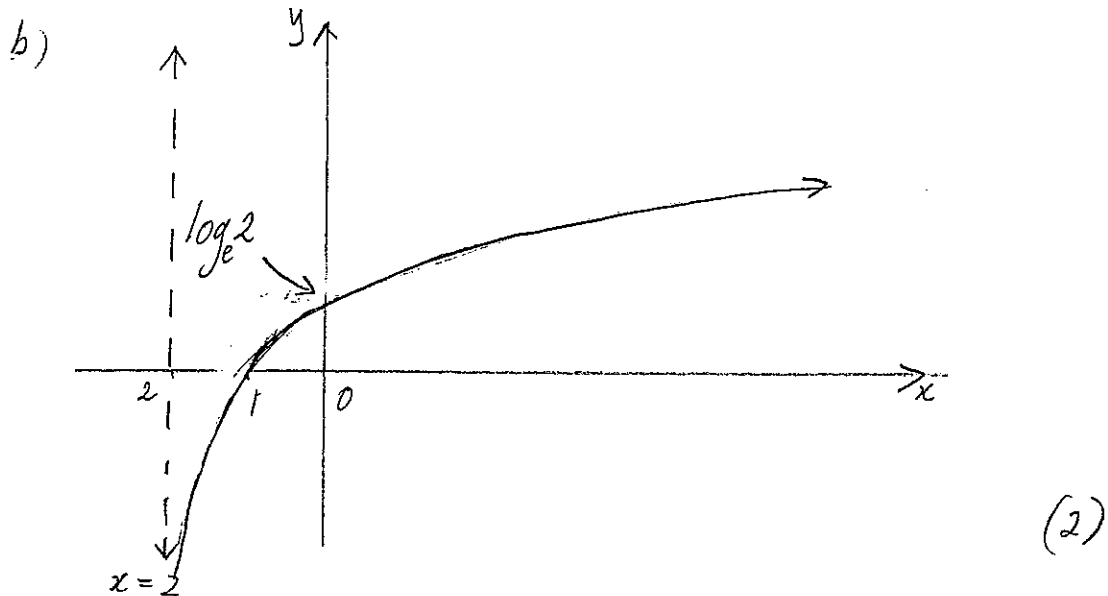
$$x = \frac{5\pi}{4}, \frac{9\pi}{4} \leftarrow \text{outside domain.}$$

$$\therefore \text{When } x = \frac{5\pi}{4} \quad (1)$$

Question 4

$$a) \quad \log_7 43700 = \frac{\ln 43700}{\ln 7} \quad (1)$$

=



c) $f(x) = x - 3 \ln x$

$$f'(x) = 1 - \frac{3}{x}$$

(1)

$$= 1 - 3x^{-1}$$

$$f''(x) = 3x^{-2}$$

$$= \frac{3}{x^2} > 0$$

for all x .

Always concave up

(1)

For stationary pts

$$f''(x) = 0$$

$$\therefore 1 - \frac{3}{x} = 0$$

$$x = 3, \quad y = 3 - 3 \ln 3$$

$\therefore (3, 3 - 3 \ln 3)$ is a minimum turning pt.

(1)

d)

$$y = xe^{-x}$$

$$\begin{aligned}\frac{dy}{dx} &= e^{-x} \cdot 1 + x \cdot (-e^{-x}) \\ &= e^{-x} - xe^{-x}\end{aligned}\quad (1)$$

$$\begin{aligned}\frac{d^2y}{dx^2} &= -e^{-x} - [e^{-x} - xe^{-x}] \\ &= -2e^{-x} + xe^{-x} \\ &= e^{-x}(x-2) \\ &= 0 \quad \text{when } x=2\end{aligned}\quad (2)$$

$\therefore$ Possible pt of inflexion at $x=2$.

$$\text{When } x < 2, \quad \frac{d^2y}{dx^2} = (+)(-) < 0$$

$$\text{when } x > 2, \quad \frac{d^2y}{dx^2} = (+)(+) > 0 \quad (1)$$

$\therefore$ Concavity Change

$\therefore$ Pt. of Inflexion at $x=2$

Question 5

$$a) \quad f'(x) = \cos(2x) - \sec^2 x$$

$$f(x) = \frac{1}{2} \sin(2x) - \tan x + c \quad (1)$$

$$f\left(\frac{\pi}{4}\right) = \frac{1}{2} - 1 + c$$

$$\frac{1}{2} = \frac{1}{2} - 1 + c$$

$$c = 1$$

$$\therefore f(x) = \frac{1}{2} \sin(2x) - \tan x + 1 \quad (1)$$

$$b) \quad (i) \quad \cos \theta = \frac{AP}{AB}$$

$$\cos \theta = \frac{AP}{2R}$$

$$\therefore AP = 2R \cos \theta \quad (1)$$

$$(ii) \quad \text{Area} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 2R \times 2R \cos \theta \cdot \sin \theta$$

$$= 2R^2 \sin \theta \cos \theta \quad (1)$$

$$(iii) \quad A = R^2 \sin 2\theta \quad \text{using given substitution}$$

$$\frac{dA}{d\theta} = 2R^2 \cos 2\theta$$

$$= 0 \quad \text{when } \theta = \frac{\pi}{4} \quad (1)$$

$$\frac{d^2A}{d\theta^2} = -4R^2 \sin 2\theta$$

$$= -4R^2 < 0 \quad \text{when } \theta = \frac{\pi}{4} \quad (2)$$

$\therefore$ Maximum Area occurs when $\theta = \frac{\pi}{4}$.

$$c) (i) \text{ Arc ADB} = r\theta \quad \text{using } L = r\theta. \quad (1)$$

$$\text{In } \triangle OBC, \tan(180-\theta) = \frac{BC}{r}$$

$$-\tan \theta = \frac{BC}{r}$$

$$BC = -r \tan \theta \quad (1)$$

$$\therefore \text{Arc ADB} + \text{Interval BC} = r\theta - r \tan \theta.$$

$$L = r(\theta - \tan \theta)$$

$$\frac{L}{r} = \theta - \tan \theta \quad (1)$$

$$(ii) \quad \frac{1}{2} r^2 \theta = \frac{3}{8} \pi r^2 \quad (1)$$

$$\theta = \frac{3\pi}{4} \quad (1)$$

$$\begin{aligned} \therefore \frac{L}{r} &= \frac{3\pi}{4} - \tan \frac{3\pi}{4} \\ &= \frac{3\pi}{4} + 1 \end{aligned} \quad (1)$$